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Global Rice Paddy Inventory (GRPI)

Chen, Zichong; Lin, Haipeng; Balasus, Nicholas; Hardy, Andy; East, James D.; Zhang, Yuzhong; Runkle, Benjamin R. K.; Hancock, Sarah E.; Taylor, Charles A.; Du, Xinming; Sander, Bjoern Ole; Jacob, Daniel J.

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Earth's Future

RESEARCH ARTICLE

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Key Points:

- We developed a new Global Rice Paddy Inventory of methane emissions at $0.1^\circ \times 0.1^\circ$ monthly resolution using Landsat satellite inundation data
- Our global emission of $39.3 \pm 4.7 \text{ Tg a}^{-1}$ is higher than previous inventories that use outdated rice maps and IPCC-recommended emission factors now considered too low
- Countries can mitigate methane without compromising food security by developing high-yield cultivars, upland rice agriculture, water management, and organic matter management

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

Z. Chen,
zchen1@g.harvard.edu

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Author Contributions:

Conceptualization: Zichong Chen, Daniel J. Jacob

Data curation: Zichong Chen,

Haipeng Lin, Nicholas Balasus

Formal analysis: Zichong Chen,

Haipeng Lin, Nicholas Balasus

Funding acquisition: Daniel J. Jacob

Methodology: Zichong Chen,

Haipeng Lin, Andy Hardy

Supervision: Daniel J. Jacob

Writing – original draft: Zichong Chen, Daniel J. Jacob

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Global Rice Paddy Inventory (GRPI): A High-Resolution Inventory of Methane Emissions From Rice Agriculture Based on Landsat Satellite Inundation Data

Zichong Chen^{1,2} , Haipeng Lin², Nicholas Balasus² , Andy Hardy³ , James D. East² , Yuzhong Zhang⁴ , Benjamin R. K. Runkle⁵ , Sarah E. Hancock², Charles A. Taylor⁶, Xinming Du⁷, Bjoern Ole Sander⁸, and Daniel J. Jacob²

¹Hong Kong University of Science and Technology, Guangzhou, China, ²John A. Paulson School of Engineering and Applied Sciences, Harvard University, Cambridge, MA, USA, ³Department of Geography and Earth Sciences, Aberystwyth University, Aberystwyth, UK, ⁴Institute of Advanced Technology, Westlake Institute for Advanced Study, Hangzhou, China, ⁵Department of Biological & Agricultural Engineering, University of Arkansas, Fayetteville, AR, USA, ⁶John F. Kennedy School of Government, Harvard University, Cambridge, MA, USA, ⁷Department of Economics, National University of Singapore, Singapore, Singapore, ⁸International Rice Research Institute (IRRI), Los Baños, Philippines

Abstract Rice agriculture is a major source of atmospheric methane, but current emission inventories are highly uncertain, mostly due to poor rice-specific inundation data. Inversions of atmospheric methane observations can help to better quantify rice emissions but require high-resolution prior information on the location and timing of emissions. Here we use Landsat satellite data at 30 m resolution to map the global monthly distribution of rice paddy fractional areas on a $0.1^\circ \times 0.1^\circ$ ($\sim 10 \times 10 \text{ km}$) grid by optimizing an algorithm for flooded vegetation and combining it with a 30 m global cropland database and rice-specific data. We validate this global rice paddy map with an independent US rice database and with seasonal flux measurements from the FLUXNET CH_4 network, estimating errors on rice area fraction of 31% on the $0.1^\circ \times 0.1^\circ$ grid and 10% regionally. We combine the rice paddy map with an extensive global data set of emission factors (EFs) per unit of rice paddy area. The resulting Global Rice Paddy Inventory (GRPI) provides methane emission estimates at $0.1^\circ \times 0.1^\circ$ ($\sim 10 \times 10 \text{ km}$) spatial resolution and monthly resolution. Our global emission of $39.3 \pm 4.7 \text{ Tg a}^{-1}$ for 2022 (best estimate and error standard deviation) is higher than previous inventories that use outdated rice maps and IPCC-recommended EFs now considered to be too low. China is the largest rice emitter in GRPI ($8.2 \pm 1.0 \text{ Tg a}^{-1}$), followed by India ($6.5 \pm 1.0 \text{ Tg a}^{-1}$), Bangladesh ($5.7 \pm 1.2 \text{ Tg a}^{-1}$), Vietnam ($5.7 \pm 1.0 \text{ Tg a}^{-1}$), and Thailand ($4.4 \pm 0.9 \text{ Tg a}^{-1}$). These five countries together account for 78% of global total rice emissions. Seasonality of emissions varies considerably between and within individual countries reflecting differences in climate and crop practices. We define a rice methane intensity (methane emission per unit of rice produced) to assess the potential of mitigating methane emission without compromising food security. We find national methane intensities ranging from 10 to 120 kg methane per ton of rice produced (global mean 51) for major rice-growing countries. Countries can achieve low intensities with high-yield cultivars, upland rice agriculture, water management, and organic matter management.

Plain Language Summary Rice agriculture is a major source of atmospheric methane, a potent greenhouse gas with strong warming potential. Current emission estimates for rice agriculture are highly uncertain because of poor inundation data. Here we use Landsat satellite data to develop a new Global Rice Paddy Inventory (GRPI) of methane emissions at 10 km resolution for each month of 2022. We find that global rice methane emissions are higher than previously thought, at 39.3 million metric tons in 2022. Five countries (China, India, Bangladesh, Vietnam, and Thailand) account for 78% of these emissions. We introduce a metric of methane intensity - methane emitted per ton of rice produced - to assess the potential to reduce methane emission without compromising food security. We find that methane intensities vary widely between countries.

1. Introduction

Rice is a staple food for more than half of the world's population, and rice paddies cover 170 million ha or 1.3% of the global land surface. Flooded anoxic conditions combined with a high organic load and standing vegetation make rice paddies a large seasonal source of methane (CH_4), a potent greenhouse gas (US EPA, 2019).

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Rice agriculture is concentrated in East, South, and Southeast Asia. These three regions contributed 90% of global rice production in 2022 (FAOSTAT, 2024). The rest is distributed in Africa, the US, South America, and southern Europe. Flooding for rice paddies is seasonal, governed by rice cultivation cycles. Bottom-up estimates of methane emissions from rice agriculture apply EFs to rice paddy areas over the rice growing season (IPCC, 2006). Upland rice agriculture, where the fields are not flooded, emits negligible methane (Minami & Neue, 1994). Bottom-up methane emission inventories compiled by the Global Carbon Project (GCP) give global rice emissions of 25.1–37.5 Tg a^{−1} for the 2010–2019 decade, accounting for 6%–11% of total anthropogenic emissions of methane (Saunio et al., 2024). Rice emission from China, the world's largest rice producer, ranges from 5.3 to 14.2 Tg a^{−1} in these inventories. Rice is considered to have the largest methane mitigation potential in South and Southeast Asia compared to other sectors (Shindell et al., 2024).

Top-down emission estimates by inversion of satellite observations of atmospheric methane can help to improve bottom-up inventories by using these inventories as prior information (Jacob et al., 2016). High spatial resolution is required in the bottom-up inventory to enable interpretation of the observed atmospheric concentration gradients. Top-down studies generally use the Emissions Database for Global Atmospheric Research (EDGAR; Crippa et al., 2024) inventory as prior estimate because it uniquely provides 0.1° × 0.1° spatial resolution and monthly temporal resolution. But these studies found that rice emissions in EDGAR have large errors in magnitude and seasonality (Palmer et al., 2021; Yu et al., 2023), and that errors in spatial distribution compromise the ability to separate rice emissions from other sectors such as coal in southern China (Qu et al., 2021). Global inversions of satellite data using EDGAR as prior estimate found top-down estimates in the range of 40–45 Tg a^{−1} (Qu et al., 2021; Yu et al., 2023).

The principal challenge in constructing spatially and temporally resolved rice emission inventories is the availability of inundation data. Previous studies used satellites to map wetland inundation extent (Arino et al., 2012; Gerlein-Safdi et al., 2021; Lehner & Döll, 2004; Muro et al., 2018; Pekel et al., 2016) but with spatial resolutions of 300 m or larger, which is too coarse to resolve rice paddies (Inman & Lyons, 2020). They also did not differentiate between open water and flooded vegetation. Flooded vegetation is the main driver of methane emission because of the large decaying organic pools and the vascular transport of methane through the plants (Helfter et al., 2022; Kyzivat et al., 2022; Rahman & Yamamoto, 2020; Rajendran et al., 2023; Shaw et al., 2022). Landsat satellite imagery is available that maps open water and flooded vegetation at 30 m resolution (Díaz-Delgado et al., 2016). Using the Landsat data covering the visible, near-infrared (NIR) and shortwave-infrared (SWIR) bands, Hardy et al. (2020, 2023) developed the Tropical Wetland mapping tool (TropWet) that can map inundation extent at 30 m resolution, and they applied it to wetlands across Africa. Here we adapt their methods to map rice paddies worldwide.

Emission factors are another source of uncertainty in rice emission inventories. Emission factors are commonly expressed as mean methane emission per unit of rice paddy area during the growing season (IPCC, 2019). National emission inventories reported to the United Nations Framework Convention on Climate Change (UNFCCC; <https://unfccc.int/topics/mitigation/resources/registry-and-data/ghg-data-from-unfccc>) often rely on the Intergovernmental Panel on Climate Change (IPCC) Tier 1 or Tier 2 methods, where Tier 1 uses global defaults and Tier 2 uses country-specific EFs though sometimes with limited adjustments from Tier 1 (Wang et al., 2018). Previous studies have underscored the limited field validation and biases in the IPCC-recommended EFs (Chen et al., 2024; Ouyang et al., 2023; Reavis et al., 2023). Nikolaisen et al. (2023) compiled a global data set from 2301 field measurements and developed a generalized additive model to infer country-specific EFs as a function of local agricultural practices and environmental factors (soil texture, planting method, rice cultivar, water regime, and climate classification).

Here we use Landsat satellite data combined with rice-specific data to produce a new global gridded map of rice paddies at (0.1° × 0.1°) spatial resolution and monthly temporal resolution, and we combine this map with national EFs from Nikolaisen et al. (2023) to produce the Global Rice Paddy Inventory (GRPI). Global Rice Paddy Inventory provides a basis for interpreting atmospheric observations of methane, particularly from satellites, to better quantify methane emissions from rice agriculture and distinguish them from other sectors. Our inventory is for 2022 but the methods can be readily applied to other years. Interannual variability of emissions in Asia is found to be less than 8% (Ouyang et al., 2023). Decadal trends can be more important (Anand et al., 2005; Chen et al., 2024; Wang et al., 2023; Zhang et al., 2022).

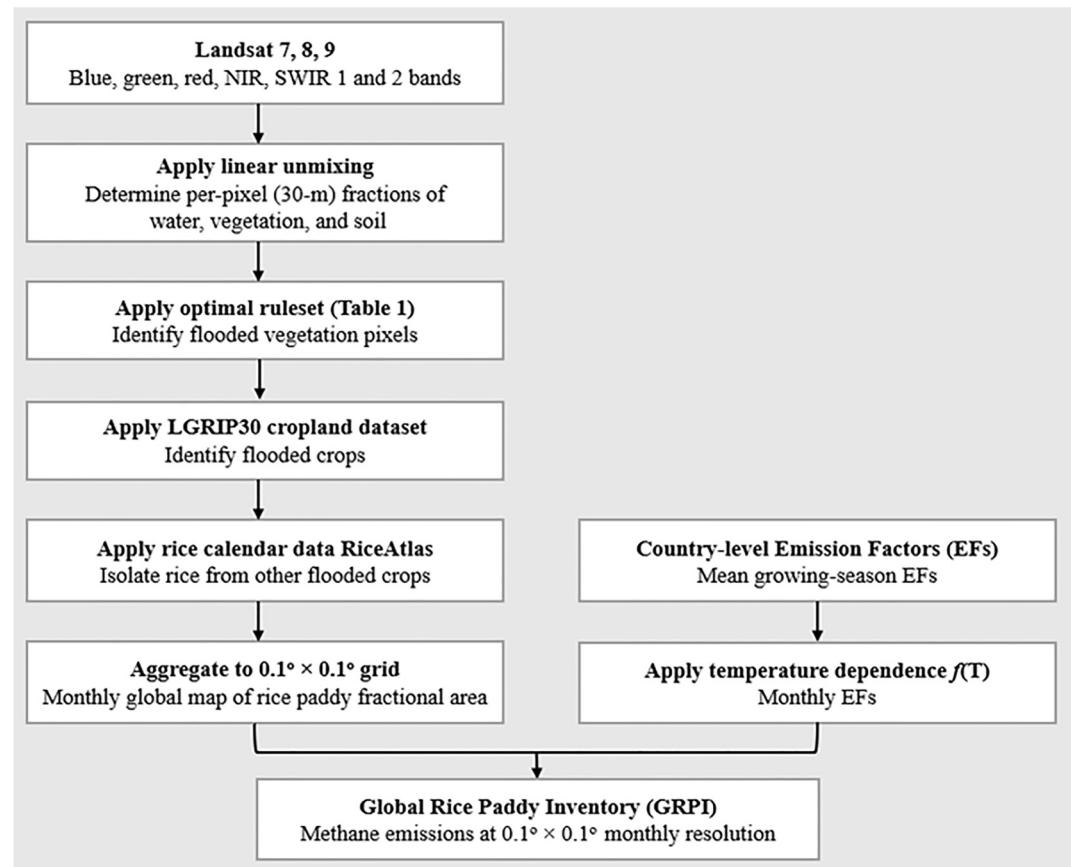


Figure 1. Flowchart for generating the Global Rice Paddy Inventory of methane emissions from rice agriculture. Near-infrared (NIR) and shortwave-infrared are NIR and short-wave infrared bands respectively. LGRIP30 is the global Landsat-derived rainfed- and irrigated-cropland product at 30 m resolution (Teluguntla et al., 2023).

2. Data and Methods

Figure 1 shows the general schematic for constructing GRPI. We use Landsat imagery for flooded vegetation combined with global cropland data sets to construct a global monthly $0.1^\circ \times 0.1^\circ$ ($\sim 10 \times 10$ km) fractional area map for rice paddy inundation (Sect 2.1). This is then combined with country-specific EFs from Nikolaisen et al. (2023) including temperature dependence (Sect. 2.2).

2.1. Global Mapping of Rice Paddies

We start by applying the TropWet Google Earth Engine (GEE)-based automatic mapping system developed by Hardy et al. (2020, 2023) to generate global monthly data on inundated vegetation from Landsat 7, 8, and 9 surface reflectance data at 30 m pixel resolution using six spectral bands including blue (0.45–0.51 μm), green (0.53–0.59 μm), red (0.64–0.67 μm), NIR (0.85–0.88 μm), SWIR1 (1.57–1.65 μm), and SWIR2 (2.11–2.29 μm). We collect images from Landsat 7, 8, and 9 over a month and combine them into one monthly composite image per 30 m pixel by taking the median value of the pixel surface reflectance in each band; previous studies have shown that median compositing effectively reduces noise from transient cloud and haze contamination by filtering out extreme values (Azzari & Lobell, 2017; Qiu et al., 2023; Zhu et al., 2015). We then determine the per-pixel fractions of water, vegetation, and bare soil by applying linear spectral unmixing to the composite Landsat imagery. This technique is widely used to estimate the proportion of different land cover types (endmembers) within a pixel by modeling its reflectance as a linear combination of the spectral signatures of those endmembers (Gevaert & García-Haro, 2015). We also calculate the normalized difference water index (NDWI) from the NIR and SWIR reflectance as $\text{NDWI} = (\text{NIR} - \text{SWIR1}) / (\text{NIR} + \text{SWIR1})$, a measure of the water content in leaves of plants. Two terrain metrics including slope angle and Height Above Nearest Drainage (HAND), both

Table 1
Optimal Ruleset Used for Identification and Classification of Inundated Landsat Pixels^a

Pixel classification	Parameters					
	Water fraction	Vegetation fraction	Bare soil fraction	NDWI ^b	Slope ^c	HAND ^c
Open water	>75%				<2.78%	<30 m
Emergent vegetation	25%–75%	25%–75%			<1.67%	
Aquatic vegetation		>75%		>NDWI _{p70}	<1.67%	<15 m
Wet soil	25%–75%		25%–75%		<2.78%	

^aParameters from Hardy et al. (2023) with values optimized for rice paddy identification (see text). Emergent vegetation and aquatic vegetation categories (referred to together as flooded vegetation) are potential rice paddies if further associated with rice cultivation (see Figure 1 and text). ^bNDWI derived from the Landsat NIR and SWIR radiances as $NDWI = (NIR - SWIR1)/(NIR + SWIR1)$. NDWI_{p70} is the 70th percentile of the NDWI values across pixels within the $0.1^\circ \times 0.1^\circ$ grid cell for each month. ^cTerrain slope and HAND are from the 30 m SRTM DSM.

generated using the 30 m Shuttle Radar Topography Mission (SRTM) Digital Surface Model (DSM) available in GEE (Farr et al., 2007), are used to further discriminate inundated areas from dry areas. Hardy et al. (2023) used this ensemble of six parameters to produce a set of logical rules (ruleset) identifying inundated 30 m pixels and classifying them as open water, emergent vegetation (herbaceous vegetation emerging from water), aquatic vegetation (floating vegetation completely obscuring the water below), or wet soil. We use the ruleset from Hardy et al. (2023) as a first guess pixel classification map. The optimal ruleset given in Table 1 is refined for rice paddies as described below. We combine emergent and aquatic vegetation (referred to hereafter as flooded vegetation) as potential rice paddies.

Our next step is to separate flooded crops from natural wetlands in the flooded vegetation category. We use for that purpose the global Landsat-derived rainfed- and irrigated-cropland product (LGRIP30 version 1.2; Teluguntla et al., 2023) at the same 30 m resolution. We further isolate rice paddies from other flooded crops such as taro and cranberries by using the global RiceAtlas data set (Laborte et al., 2017), which gives rice area in 2,725 spatial units ranging from 11 to 129 kha depending on country. This would not account for recent changes in rice cultivation patterns, but other flooded crops amount to only 1.4% of the rice area and do not tend to be substituted for rice.

The 30 m Landsat resolution is important for identifying rice paddy locations but there is substantial noise at that resolution. To reduce this noise, we compile our rice paddy product as a spatially averaged monthly rice paddy area fraction on a $0.1^\circ \times 0.1^\circ$ grid. The $0.1^\circ \times 0.1^\circ$ resolution is commensurate with gridded bottom-up emission inventories from other sectors used in inversions of methane observations (Maasakkers et al., 2023; Scarpelli et al., 2022) including EDGAR. Each monthly $0.1^\circ \times 0.1^\circ$ grid cell averages from 439 to 138,384 (mean of 80,410) individual Landsat clear-sky scenes. We then compare the result to the US Department of Agriculture (USDA) Cropland Data Layer (CDL) (Li et al., 2024) for California, which provides on-the-ground annual rice location data at 30 m resolution and with high accuracy (Luman & Tweddle, 2008), and iteratively adjust the ruleset parameters at $0.1^\circ \times 0.1^\circ$ grid resolution to achieve an optimal match. Table 1 lists the optimized parameters. Calibrating the ruleset for California rice paddies may induce errors when applied worldwide. We estimate this error in Sect. 2.3 by independent evaluation with the CDL data for the South-Central US (the other and more extensive rice-growing area in the US) and by comparison to FLUXNET-CH₄ tower flux measurements in the US, Korea, and Japan.

2.2. Emission Factors

We use country-specific EFs from Nikolaisen et al. (2023) to compute rice emissions per unit of rice paddy area during the rice growing season. Nikolaisen et al. (2023) compiled 2,301 rice paddy field measurements globally and developed a generalized additive model to estimate local mean EFs over the rice growing season as a function of soil texture, pre-season water status, growing-season water regime, planting method, rice cultivar, organic amendment, and climate zone. They then applied that model to derive mean EFs for individual countries, given in Table 2. The highest EF is for Vietnam ($3.60 \text{ kg ha}^{-1} \text{ d}^{-1}$) due to widespread application of farmyard manure and continuous flooding (Pandey et al., 2014; Vo et al., 2018, 2020). The lowest EFs are for India ($0.95 \text{ kg ha}^{-1} \text{ d}^{-1}$)

Table 2
Rice Paddy Emission Factors for Methane in Individual Countries and Regions^a

	EF (kg ha ⁻¹ d ⁻¹)	Relative uncertainty (%) ^b
Bangladesh	2.08	11.1
Brazil	2.13	4.2
China	2.41	2.1
Ghana	2.54	20.0
India	0.95	5.3
Indonesia	1.96	4.6
Italy	2.21	8.1
Japan	1.53	9.8
Myanmar	1.49	24.2
Philippines	0.97	6.2
Portugal	1.36	33.1
South Korea	2.39	3.3
Spain	2.51	13.1
Thailand	1.27	9.4
Uruguay	1.49	17.4
US	1.25	4.8
Vietnam	3.60	6.7
East Asia	2.03	4.9
South Asia	1.16	6.9
Southeast Asia	1.91	5.2
Europe	2.16	7.4
Africa	2.54	20.0
North America	1.25	4.8
South America	2.04	4.9
World	1.97	2.1

^aFrom Nikolaisen et al. (2023). ^bFrom the residual standard deviations reported by Nikolaisen et al. (2023) in fitting their generalized additive model to individual measurements in the different countries.

due to sandy soils and rice-wheat rotation systems (Nikolaisen et al., 2023), and for the Philippines (0.97 kg ha⁻¹ d⁻¹) due to its adoption of dry direct seeding (seeds are sown directly into dry soils) and alternate wetting and drying practices of rice paddies (Government of Philippines, 2014).

Nikolaisen et al. (2023) derived a global mean EF of 1.97 kg ha⁻¹ d⁻¹, whereas the IPCC (2006) Tier 1 recommendation is 1.30 kg ha⁻¹ d⁻¹, which is used in EDGARv8. Older inventories have also used lower EFs (Wang et al., 2018; Yan et al., 2005). The Nikolaisen et al. (2023) values are based on considerably more information. For Vietnam, for example, they use data from 36 field sites covering all major rice regions whereas the Wang et al. (2018) model was based on only two sites. Nikolaisen et al. (2023) also take into account the effects of soil texture, planting method, and cultivar type, information that is missing from previous inventories.

These EFs may not fully represent total emissions from the whole rice-water system, as chamber-based measurements can omit plant-stem-mediated emissions depending on placement and coverage (Minamikawa et al., 2015), while in-paddy measurements may overlook emissions from surrounding drainage ditches and irrigation channels (Martínez-Eixarch et al., 2018). Additionally, these EFs do not reflect most recent changes in rice paddy patterns, as many field measurements were conducted before 2020. However, the interannual variability of rice emissions is found to be small (Ouyang et al., 2023).

The EFs from Nikolaisen et al. (2023) do not include temperature dependence, which may be important for emission seasonality. We add a temperature dependence by applying a multiplicative monthly factor $f(T) = \max\left(1, 2.5^{\frac{T-30}{10}}\right)$ where T is monthly mean soil temperature (°C) from the Climatic Research Unit and Japanese reanalysis data set (CRU JRA v2.5; Harris, 2024), following the Dynamic Land Ecosystem Model (DLEM; Tian et al., 2010). The resulting monthly EF values are then rescaled to ensure that the annual emissions remain consistent with those calculated using the mean growing-season values from Nikolaisen et al. (2023).

2.3. Error Analysis

We estimate the errors on GRPI methane emissions by adding in quadrature the error variances on rice paddy areas and EFs. Figure 2 shows the spatial distribution of rice paddies over the South-Central US in GRPI compared to the USDA CDL data set in 2022 (Li et al., 2024). The rice fraction on the 0.1° × 0.1° grid is defined in the CDL as the maximum monthly fraction over the course of the year and we follow this same definition in GRPI for comparison. The CDL is considered to be highly reliable because of its use of on-the-ground information, including field survey data and farmer-reported acreage from the USDA Farm Service Agency (USDA NASS, 2025). The GRPI accurately identifies rice paddy locations including intensive rice-growing regions in East Arkansas and South Louisiana. Our error standard deviation on the rice paddy area fractions, expressed as one standard deviation of residuals between our product and the CDL data on the 0.1° × 0.1° grid, is 31%. We estimate a total rice area of 763 kha across the South-Central US, consistent with but 9% lower than the CDL best estimate of 841 ± 92 kha (Luman & Tweddle, 2008); from this comparison we estimate an error of 10% for our regional rice area estimates. In this manner, we estimate the GRPI error in identifying fractional rice paddy areas as 31% on the 0.1° × 0.1° grid and 10% for national to global scales.

Nikolaisen et al. (2023) reported uncertainties on their EF estimates in individual countries as one standard deviation of residuals between their generalized additive model and individual observations. We convert them to relative uncertainties (Table 2) and assume that they apply uniformly to all grid cells within an individual country

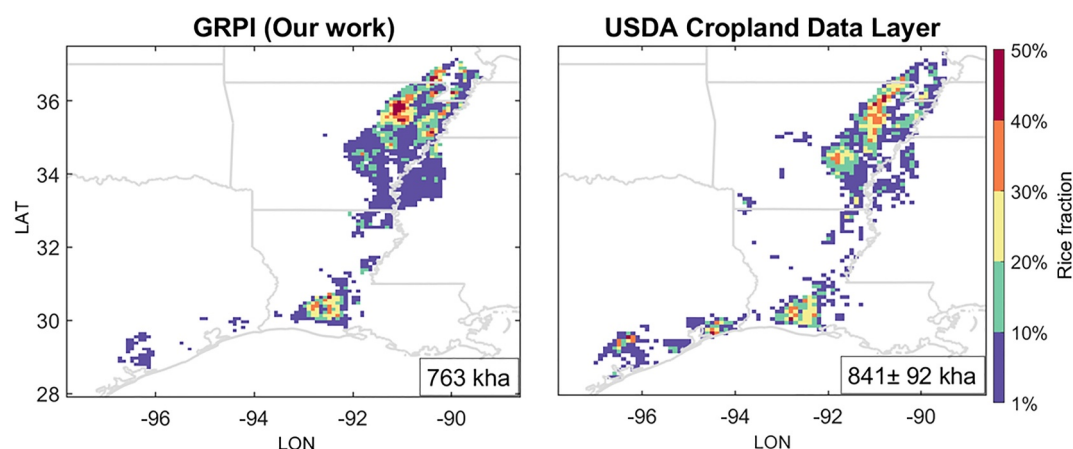


Figure 2. Rice paddies in the South-Central US in 2022. The spatial distribution in the Global Rice Paddy Inventory (our work) is compared to the Cropland Data Layer (CDL) of the US Department of Agriculture. The Figure shows the rice paddy fractional areas in $0.1^\circ \times 0.1^\circ$ grid cells. The total rice area in the region is given inset, and the uncertainty in the right panel is estimated from a 11% error assigned to total rice cropland area in the CDL (Luman & Tweddale, 2008).

as well as to the national totals. Nikolaisen et al. (2023) also reported uncertainties for their regional and global mean EFs (Table 2) and we use these values to estimate the GRPI errors for the corresponding totals.

We evaluate the GRPI methane emissions and their seasonality including error estimates by comparing to eddy flux measurements at rice paddy sites from the FLUXNET-CH₄ network (Delwiche et al., 2021). There are six such sites but three of them (IT-Cas, Italy; PH-RiF, Philippines; US-Twt, US) have highly inconsistent seasonality from year to year, likely influenced by experimental manipulations (Knox et al., 2016). Figure 3 shows results for the three remaining sites in the US, Korea, and Japan. Close agreement is found for the US-HRA/HRC (US) and JP-MSE (Japan) sites in both magnitude and seasonality. For the KR-CRK (Korea) site, we capture the observed bimodal distribution driven by mid-season drainage followed by intermittent irrigation (Hwang et al., 2020), but the observations show methane emission extending beyond the rice season which could be due to adjacent wetlands. We conclude that the GRPI seasonality is correct to the extent that it can be independently evaluated. We do not use chamber-based measurements for comparison as they are too localized (Minamikawa et al., 2015).

3. Results and Discussion

3.1. Global Spatial and Seasonal Distributions of Emissions

Figure 4 shows the global annual mean GRPI rice emissions for 2022, totaling $39.3 \pm 4.7 \text{ Tg a}^{-1}$. Five countries account for 78% of global rice emissions including China ($8.2 \pm 1.0 \text{ Tg a}^{-1}$), India ($6.5 \pm 1.0 \text{ Tg a}^{-1}$), Bangladesh ($5.7 \pm 1.2 \text{ Tg a}^{-1}$), Vietnam ($5.7 \pm 1.0 \text{ Tg a}^{-1}$), and Thailand ($4.4 \pm 0.9 \text{ Tg a}^{-1}$). The US and Brazil contribute 1.2 ± 0.2 and $0.9 \pm 0.1 \text{ Tg a}^{-1}$ respectively. Italy and Spain contribute 0.17 ± 0.03 and $0.06 \pm 0.01 \text{ Tg a}^{-1}$ respectively. Egypt contributes $0.23 \pm 0.07 \text{ Tg a}^{-1}$. Rice production in Sub-Saharan Africa has been rising rapidly (Chen et al., 2024), and Nigeria and Madagascar are the top emitters there contributing 0.66 ± 0.20 and $0.26 \pm 0.08 \text{ Tg a}^{-1}$ respectively as of 2022.

Figure 5 further shows the monthly emissions for Asia on the $0.1^\circ \times 0.1^\circ$ grid. The seasonality varies considerably with location which reflects differences in climate, water management, and rice cropping systems. In regions with single rice cropping, rice emissions typically persist for five to six months, peaking in July–August over East Asia (northern China, Japan, Korea) and in September–November over South and Southeast Asia (western India, eastern Thailand). Subtropical and tropical regions more commonly practice double or triple cropping, where rice emissions can last for eight months or longer, with seasonality further complicated by local management practices (mid-season drainage, intermittent irrigation; Qian et al., 2023).

3.2. Comparison With Previous Inventories

Figure 6 compares our GRPI for 2022 to previous inventories. The GCP values (from five different bottom-up inventories) are for 2008–2017, and the UNFCCC inventories are from the most recent national reports

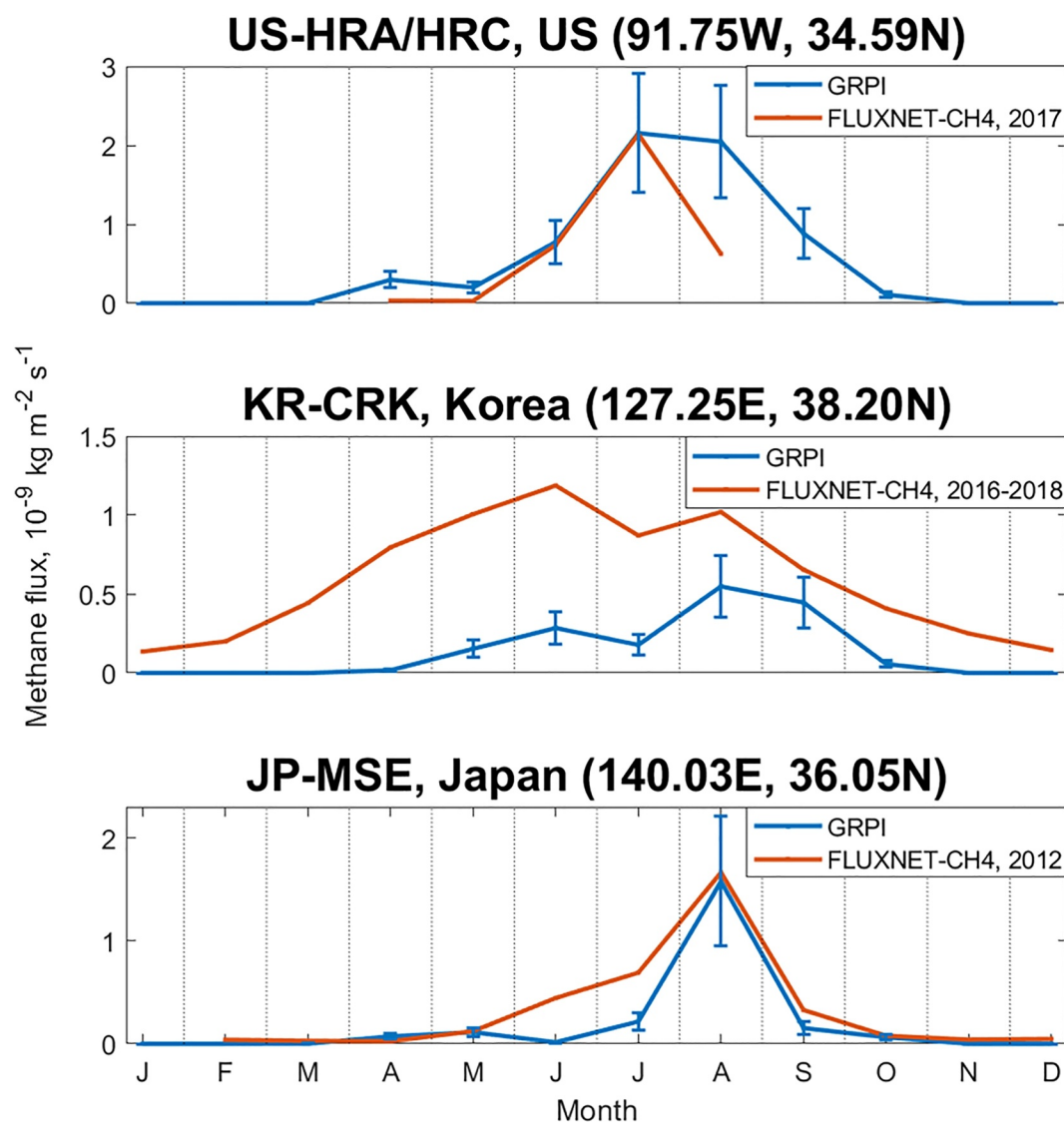


Figure 3. Seasonality of methane emissions at rice paddy sites in the FLUXNET-CH₄ network. Eddy flux measurements at the sites in different years are compared to our Global Rice Paddy Inventory (GRPI) inventory for 2022 on the $0.1^\circ \times 0.1^\circ$ grid. Vertical bars indicate GRPI error standard deviations on that grid. The red line in the top panel shows the mean values of two neighboring HRA and HRC sites in Arkansas, US.

(<https://unfccc.int/topics/mitigation/resources/registry-and-data/ghg-data-from-unfccc>; Table S1 in Supporting Information S1). Our global emission estimate of $39.3 \pm 4.7 \text{ Tg a}^{-1}$ is higher than the $24.6\text{--}37.7 \text{ Tg a}^{-1}$ range from the GCP and 24.8 Tg a^{-1} from the sum of UNFCCC national reports. Our difference with the GCP is mainly driven by South Asia (India and Bangladesh, discussed in detail in Sect. 3.3), for which the GCP inventories used outdated rice maps for 2000 (Ito et al., 2022; Monfreda et al., 2008; Portmann et al., 2010). Rice production increased by 54% in India and 52% in Bangladesh from 2000 to 2022 (FAOSTAT, 2024), a combination of expanded rice area and increased fertilizer use (Bora, 2022; Kamuruzzaman et al., 2024).

EDGAR is the only previous rice inventory available with fine spatial resolution and as such is widely used in inversions of atmospheric methane observations. Global rice emissions in EDGARv8 for 2022 (35.2 Tg a^{-1}) are similar to GRPI, but are double the GRPI values for China and half for South Asia (Figure 6). EDGARv8 spreads emissions over all agricultural land for many countries in South Asia (India, Bangladesh), Southeast Asia (Thailand, Vietnam), and East Asia (China, Korea, Japan). The seasonality in EDGARv8 is also uniform within individual countries. For example, rice emissions in EDGARv8 peak in June everywhere over China, in July over

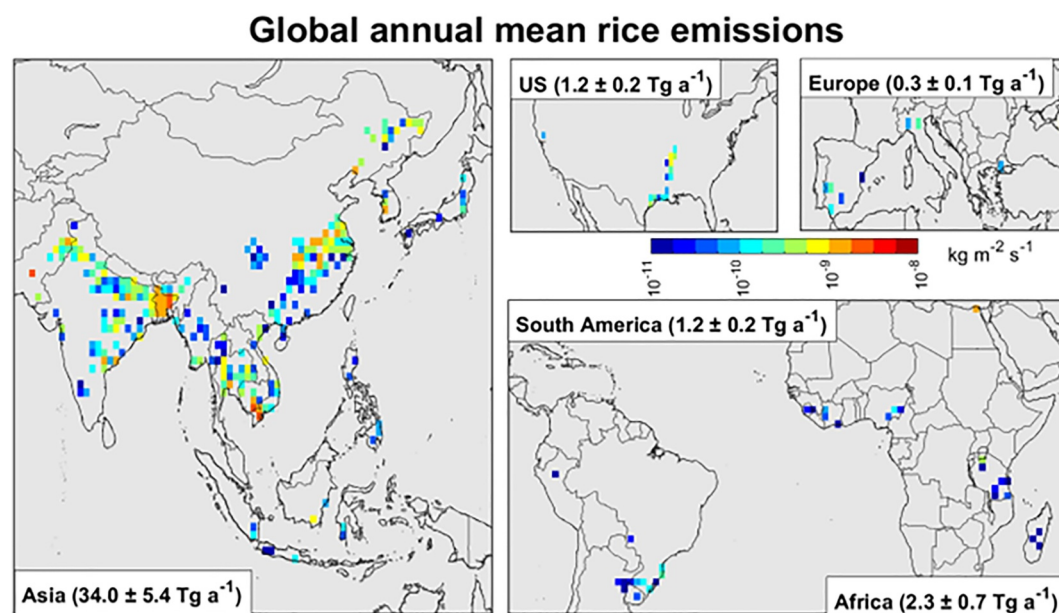


Figure 4. Annual mean rice emissions in 2022 from Global Rice Paddy Inventory in rice-producing regions. The original data at $0.1^\circ \times 0.1^\circ$ resolution are shown as $1.0^\circ \times 1.0^\circ$ averages for visibility. Gray areas have emissions lower than $1 \times 10^{-11} \text{ kg m}^{-2} \text{ s}^{-1}$. Total emissions with error standard deviations for each region are given inset. National emissions from major rice-growing countries are given in the text.

Korea and Japan, and in August over India and Bangladesh, whereas the seasonality in GRPI is more complex (Figure 5). Inversions of satellite observations of atmospheric methane using EDGAR as prior estimates found EDGAR to be too low in South and Southeast Asia, with incorrect seasonality (Palmer et al., 2021; Yu et al., 2023). Global Rice Paddy Inventory is more consistent with these inversion results.

3.3. Emissions in Major Rice-Producing Countries

Figure 7 shows monthly emissions for the top five emitting countries including China, India, Bangladesh, Vietnam, and Thailand. We discuss individual countries in what follows.

Rice emissions in China are estimated at $8.2 \pm 1.0 \text{ Tg a}^{-1}$ in GRPI, which agrees well with the GCP mean and with the UNFCCC report, but is a factor of two lower than EDGARv8 (Figure 6). EDGARv8 used an outdated rice map for 2000, ignoring the changes in irrigation, organic manure use, and rotational patterns, as well as the northward shift of rice agriculture in China in the recent two decades (Chen et al., 2013; Saunio et al., 2020). These changes have decreased rice emissions (Ouyang et al., 2023; Peng et al., 2016). Chinese emissions generally peak in July–September (Figure 5), but vary in emission season lengths largely reflecting the diverse rice cropping systems across the country. Again EFs from Nikolaisen et al. (2023) may not fully represent the total emissions from the whole rice-water system, as they capture in-paddy sources but might overlook surrounding drainage and channels (Sect. 2.2). This limitation might explain our lower estimate here compared to $10.7\text{--}12.7 \text{ Tg a}^{-1}$ from a recent regional inversion over China (Chen et al., 2022). Other satellite inversions (Liang et al., 2023; Worden et al., 2022) found much higher emissions of $15.3\text{--}29.6 \text{ Tg a}^{-1}$, reflecting their use of EDGAR as prior estimate which is likely too high.

Rice emissions in India are estimated at $6.5 \pm 1.0 \text{ Tg a}^{-1}$ in GRPI, higher than 3.4 Tg a^{-1} in the UNFCCC report and 4.1 Tg a^{-1} in EDGARv8 (Figure 6). There has been rapid development of rice agriculture in India, with national rice production increasing by 54% from 2000 to 2022 (FAOSTAT, 2024). Previous inventories use outdated rice maps for 2000 and thus underestimate rice emissions. Unlike China, where rice farming is dominated by irrigated paddies, India relies largely on rainfall (Devendra, 2012). The Indian summer monsoon, which lasts from June to September and contributes 80% of annual rainfall (Rajendran et al., 2022), dictates the rice growing season. Fields are typically sown in June–July and harvested in November, resulting in high emissions during July–October and low emissions the rest of the year.

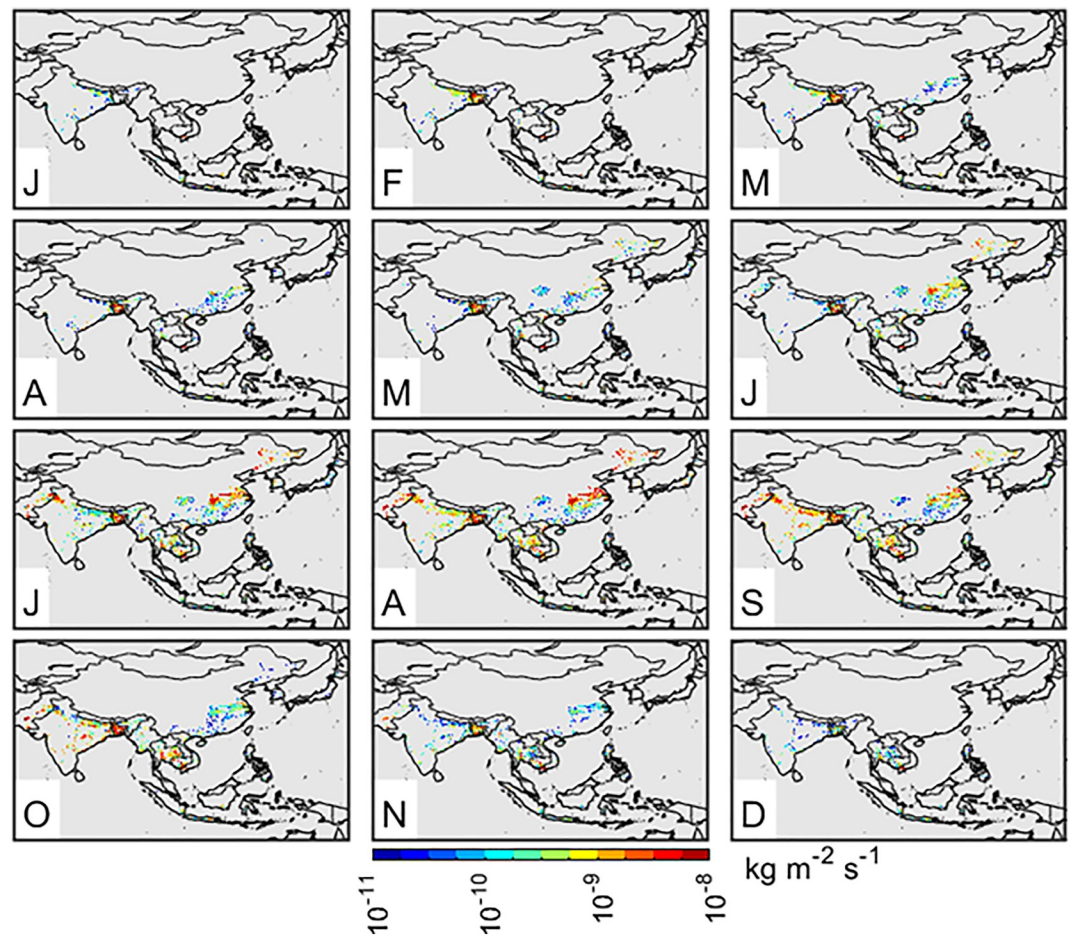


Figure 5. Monthly Global Rice Paddy Inventory rice emissions in Asia at $0.1^\circ \times 0.1^\circ$ grid resolution. Gray areas have emissions lower than $1 \times 10^{-11} \text{ kg m}^{-2} \text{ s}^{-1}$. Annual mean emissions are shown in Figure 4, and the seasonalities of national emissions for the top emitting countries are shown in Figure 7.

Unlike India, Bangladesh shows sustained emissions throughout the year except for December–January. This reflects its widespread multiple rice cropping. Bangladesh's rice emissions are estimated at $5.7 \pm 1.2 \text{ Tg a}^{-1}$ in GRPI, higher than 0.37 Tg a^{-1} in the UNFCCC report and 2.3 Tg a^{-1} in EDGARv8. Again, previous inventories use outdated rice maps for 2000 ignoring the rapid development of rice agriculture in Bangladesh (52% increase in rice production from 2000 to 2022). The EF of $2.01 \text{ kg ha}^{-1} \text{ d}^{-1}$ (Table 2) is also higher than the IPCC (2006) Tier 1 value of $1.30 \text{ kg ha}^{-1} \text{ d}^{-1}$. Recent adoption of modern hybrid rice varieties has doubled emissions compared to local varieties because of longer growth period and intensive irrigation practices (Saha et al., 2022). The disparity in emission seasonality between India and Bangladesh, two neighboring South Asian countries with comparable climatic conditions, illustrates the impact of local agricultural practices on rice emission patterns.

Rice emissions in Vietnam are estimated at $5.7 \pm 1.0 \text{ Tg a}^{-1}$, evenly distributed over the year and higher than 1.8 Tg a^{-1} in the UNFCCC report and 1.7 Tg a^{-1} in EDGARv8. This higher estimate is largely due to an EF of $3.6 \text{ kg ha}^{-1} \text{ d}^{-1}$ which is the highest in the world (Table 2). A meta-analysis of field studies across North, Central, and South Vietnam (Vo et al., 2020) found an EF more than double the IPCC default, attributed in part to the well-established irrigation system ensuring consistent water supply to rice paddies. For example, the use of high dikes has facilitated multiple rice crops per year in the Mekong Delta floodplain, an intensive rice region in the country (Tran et al., 2018).

We estimate elevated rice emissions in Thailand from July to November, peaking in October (Figure 7). This emission seasonality aligns with the monthly normalized difference vegetation index (NDVI) data derived from the Moderate Resolution Imaging Spectroradiometer (MODIS) satellite imagery. We estimate the Thai rice

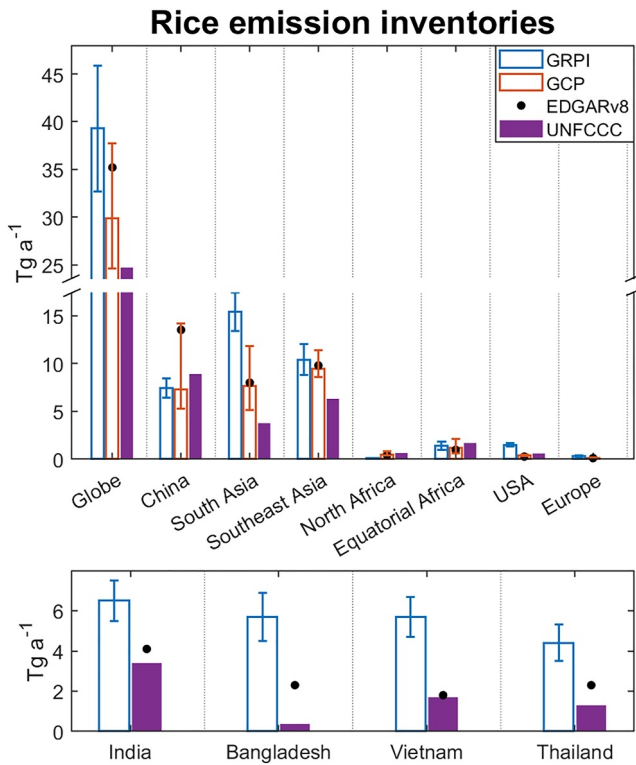


Figure 6. Comparison of our Global Rice Paddy Inventory (GRPI) inventory for 2022 to previous inventories. These include the most recent national reports to the UNFCCC, the EDGARv8 gridded emissions for 2022, and the 2008–2017 mean emissions from five bottom-up inventories contributing to the Global Carbon Project (GCP): CEDS, EDGARv432, FAO-CH₄, GAINS, and USEPA. Vertical bars for GRPI are error standard deviations. Vertical bars for GCP indicate the range for the five contributing inventories. The regions in the top panel are those resolved by GCP. The bottom panel gives comparisons between GRPI, UNFCCC, and EDGARv8 for additional countries.

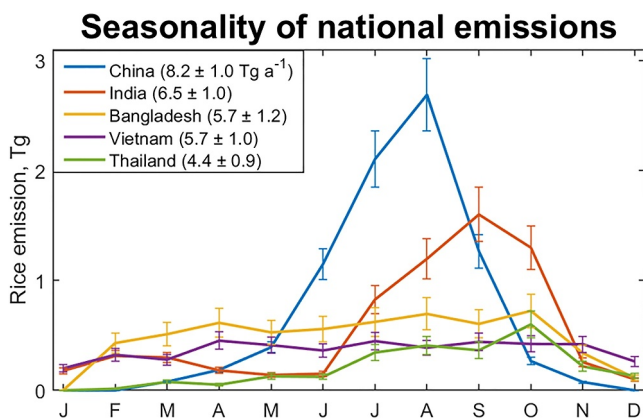


Figure 7. Monthly Global Rice Paddy Inventory (GRPI) rice emissions in 2022 for the top five emitting countries. Vertical bars indicate GRPI error standard deviations. Annual totals with error standard deviations for each country are given inset.

emissions at $4.4 \pm 0.9 \text{ Tg a}^{-1}$, higher than 2.3 Tg a^{-1} of EDGARv8, which again may be largely explained by the outdated rice map used in EDGAR ignoring the rapid rice agriculture development in Thailand (33% increase in production from 2000 to 2022; FAOSTAT, 2024).

3.4. Achieving Methane Mitigation Without Compromising Food Security

We define the metric of rice methane intensity as the methane emission per unit of rice produced, in order to assess the potential for achieving methane reduction without compromising food security in rice agriculture. Figure 8 shows methane intensities for major rice-producing countries. Values are computed from our GRPI rice emission estimates and FAOSTAT national rice production statistics (FAOSTAT, 2024). We find a wide range of methane intensities across countries, from 120 kg methane per ton of rice produced for Cambodia to 10 for the Philippines. The global mean rice methane intensity is 51 kg methane per ton of rice produced. High methane intensities reflect low yields and/or high EFs. For instance, rice yields in Cambodia, Thailand, and Pakistan are 3.5 , 3.0 , and 3.7 t ha^{-1} , lower than the global mean of 4.7 t ha^{-1} and only half of China's yield of 7.1 t ha^{-1} (FAOSTAT, 2024). Low methane-intensity countries such as India and the Philippines have low EFs of 0.95 and $0.97 \text{ kg ha}^{-1} \text{ d}^{-1}$ compared to the global default of $1.97 \text{ kg ha}^{-1} \text{ d}^{-1}$. The low intensity in India is also partly due to its upland rice agriculture (Singh et al., 2017), which produces negligible methane and is highly water-saving. Upland rice yields (typically $1\text{--}3 \text{ t ha}^{-1}$) are however much lower than lowland (paddy) rice ($4\text{--}10 \text{ t ha}^{-1}$; Liu et al., 2019; Wang et al., 2020); developing high-yield upland rice cultivars would effectively decrease methane intensity. Alternate wetting and drying irrigation of rice paddies can also reduce emissions while not affecting yields (Linguist et al., 2015; Runkle et al., 2018). Directly returning straw to paddy fields as a nutrient supply increases methane emissions, but applying straw-derived biochar (biomass charcoal) instead reduces emissions and increases yields (Dong et al., 2013; Sriphirom et al., 2021; Wang et al., 2023). More generally, countries can achieve low methane intensities with high-yield cultivars, upland rice agriculture, water management, and organic matter management (Qian et al., 2023).

4. Conclusions

We developed a new GRPI of methane emissions at $0.1^\circ \times 0.1^\circ$ ($\sim 10 \text{ km} \times 10 \text{ km}$) monthly resolution by combining Landsat satellite imagery for flooded vegetation with crop information and a large database of EFs. This inventory is suitable for use as prior information in inversions of atmospheric methane observations and to distinguish rice emissions from other sectors. Having correct prior information for the spatial and temporal distribution of emissions is essential for inversions (Yu et al., 2023), and our unique Landsat-based rice paddy data set is of considerable value for this purpose. The rice paddy dataset from GRPI can also be useful for national emission reporting in the absence of detailed on-the-ground information. With its fine spatial and temporal resolution, GRPI aids in identifying hot-spots and optimizing resources for sustainable rice farming, and is useful for policymakers to design targeted methane reduction strategies. The inventory for 2022 together with error standard deviations is publicly available as monthly $0.1^\circ \times 0.1^\circ$ files.

The major challenge in constructing rice emission inventories is the availability of inundation data. We built upon the algorithm of Hardy et al. (2020,

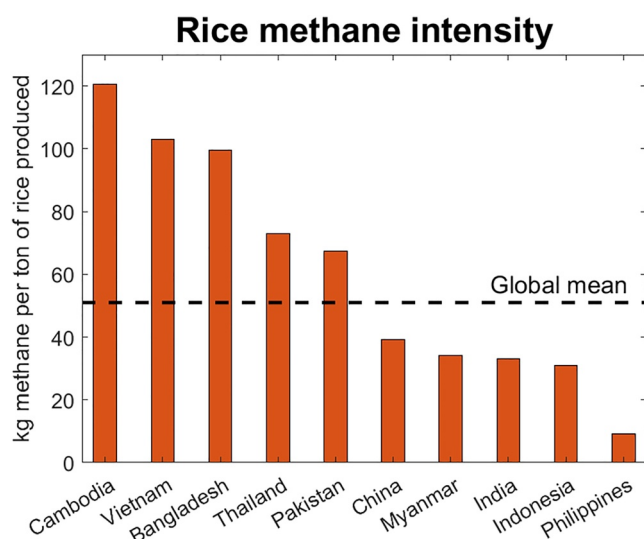


Figure 8. Rice methane intensities (methane emissions per unit of rice produced) in major rice-producing countries for 2022, in units of kg methane per ton of rice produced. Dashed horizontal line indicates the global mean methane intensity of 51 kg methane per ton of rice produced. Values are computed from our rice emission estimates and FAOSTAT country-specific rice production statistics (FAOSTAT, 2024).

(2023) using Landsat imagery to map flooded vegetation at 30 m resolution for individual months. Global cropland data sets were applied to isolate rice paddies from natural wetlands and other flooded crops. The rice paddy area maps were then re-gridded to $0.1^\circ \times 0.1^\circ$ resolution for better statistics. We applied country-specific EFs per unit of rice paddy area from the Nikolaisen et al. (2023) generalized additive model based on statistical modeling of a very large number of field measurements of rice emissions, and further applied temperature dependencies to these EFs. Evaluation of our inventory with independent rice area data for the US and with FLUXNET-CH₄ eddy flux measurements shows good agreement in emission locations, magnitude, and seasonal timing. The inventory is also consistent with spatial and seasonal distributions from top-down inverse analyses. We estimate global rice emissions in 2022 to be $39.3 \pm 4.7 \text{ Tg a}^{-1}$ (best estimate and error standard deviation), higher than national UNFCCC reports totaling 24.8 Tg a^{-1} , and the GCP's range of $24.6\text{--}37.7 \text{ Tg a}^{-1}$ for the 2008–2017 decade. The inventories contributing to the GCP used outdated 2000 rice maps and IPCC-recommended EFs now recognized to be too low. Our estimates diverge the most from GCP in South Asia where rice agriculture has developed rapidly in the past two decades. The spatially resolved EDGARv8 inventory widely used in inversions of atmospheric observations has a global emissions of 35.2 Tg a^{-1} , comparable to GRPI, but double for China and half for South Asia, again it seems because of an outdated rice map. The EDGARv8 seasonality is uniform within a given country and we find this to be a significant source of error.

China is the highest rice emitter in GRPI ($8.2 \pm 1.0 \text{ Tg a}^{-1}$), followed by India ($6.5 \pm 1.0 \text{ Tg a}^{-1}$), Bangladesh ($5.7 \pm 1.2 \text{ Tg a}^{-1}$), Vietnam ($5.7 \pm 1.0 \text{ Tg a}^{-1}$), and Thailand ($4.4 \pm 0.9 \text{ Tg a}^{-1}$). These five countries together account for 78% of global emissions. Rice emissions in China generally peak in July–August, but differ in emission season lengths due to diverse rice cropping systems. Rice emissions in India are high from July to October, dictated by the summer monsoon. Despite sharing similar climatic conditions with India, Bangladesh has high rice emissions through most of the year due to widespread multiple rice cropping practices.

The potential for reducing methane emissions from rice agriculture while not compromising food security can be quantified using a rice methane intensity metric defined as methane emission per unit of rice produced. Methane intensities in major rice-producing countries vary by one order of magnitude from 10 to 120 kg methane per ton of rice produced, with a global mean of 51. High methane intensities reflect low yields and/or high EFs. High methane-intensity countries including Cambodia, Thailand, and Pakistan have lower-than-average rice yields. Low methane-intensity countries including India and the Philippines have low EFs. Developing high-yield cultivars, upland rice agriculture, water management, and organic matter management are effective avenues for countries to achieve low methane intensities.

The EFs of Nikolaisen et al. (2023) may not fully represent total emissions from the whole rice-water system, as chamber-based measurements can be too localized and some measurements may only capture in-paddy sources but overlook surrounding drainage ditches and channels. Also, validation remains limited due to sparse ground-based methane flux measurements, highlighting the need for expanded field observations, particularly in Africa and South America.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Eddy flux measurement data in the FLUXNET-CH₄ network is available at <https://fluxnet.org/data/fluxnet-ch4-community-product/>. Methane emission estimates from the Global Carbon Project are available at Friedlingstein et al. (2020). The Food and Agriculture Organization database (FAOSTAT) is available at <https://www.fao.org/>

faostat/en/#data/QCL. The Landsat satellite imagery and algorithms to support the derivation of inundation extent is openly accessible at Chen (2025). Our gridded $0.1^{\circ} \times 0.1^{\circ}$ GRPI inventory data is publicly available at Chen and Jacob (2025).

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